RETRIEVAL-AUGMENTED ACADEMIC ASSISTANT WITH LOCAL RAG IMPLEMENTATION

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ABSTRACT

The increasing volume of scientific publications has made it challenging for researchers to efficiently retrieve relevant information using conventional academic search systems. Retrieval-Augmented Generation (RAG) addresses this challenge by combining information retrieval with large language models to generate evidence-grounded responses. However, existing RAG systems are primarily cloud-based, with limited research on locally deployed implementations using open-weight models.

This study proposes and evaluates a locally deployed RAG pipeline for academic question answering. The proposed framework integrates query expansion, hybrid retrieval, cross-encoder re-ranking, and local language models. Three experiments were conducted to select the most suitable retriever and generator models, optimize the pipeline through hyperparameter tuning and QLoRA fine-tuning, and evaluate its performance using automatic evaluation metrics.

The results identified **granite-embedding:30m** as the best-performing retriever and **Qwen2.5:3B** as the most suitable generator. After optimization, the proposed pipeline achieved a **19.3% improvement** in Faithfulness while maintaining high Answer Relevancy and improving Context Recall, resulting in an overall improvement in the RAGAS Index.

Overall, this study demonstrates that lightweight open-weight models can effectively support locally deployed academic question-answering systems and provides a reproducible framework for selecting, optimizing, and evaluating RAG components using automatic evaluation methods.

Keywords: Retrieval-Augmented Generation (RAG), Academic Question Answering, Large Language Models (LLMs), Local Deployment, Automatic Evaluation.

ABSTRAK

Peningkatan jumlah penerbitan saintifik telah menjadikan proses mendapatkan maklumat yang relevan semakin mencabar menggunakan sistem carian akademik konvensional. Retrieval-Augmented Generation (RAG) menangani cabaran ini dengan menggabungkan pencarian maklumat dan model bahasa berskala besar bagi menghasilkan jawapan yang berasaskan bukti. Walau bagaimanapun, kebanyakan sistem RAG sedia ada bergantung pada perkhidmatan awan, manakala penyelidikan berkaitan pelaksanaan tempatan menggunakan model sumber terbuka masih terhad.

Kajian ini mencadangkan dan menilai satu saluran RAG yang dilaksanakan secara tempatan untuk tugasan soal jawab akademik. Kerangka yang dicadangkan mengintegrasikan pengembangan pertanyaan, pencarian hibrid, penyusunan semula menggunakan cross-encoder, dan model bahasa tempatan. Tiga eksperimen telah dijalankan untuk memilih model pencari dan penjana yang paling sesuai, mengoptimumkan saluran melalui penalaan hiperparameter dan penalaan halus QLoRA, serta menilai prestasinya menggunakan metrik penilaian automatik.

Hasil kajian mengenal pasti granite-embedding:30m sebagai model pencari terbaik dan Qwen2.5:3B sebagai model penjana yang paling sesuai. Selepas proses pengoptimuman, saluran yang dicadangkan mencapai peningkatan 19.3% dalam metrik Faithfulness sambil mengekalkan Answer Relevancy yang tinggi dan meningkatkan Context Recall, sekali gus meningkatkan keseluruhan RAGAS Index.

Secara keseluruhannya, kajian ini menunjukkan bahawa model sumber terbuka bersaiz ringan mampu menyokong sistem soal jawab akademik yang dilaksanakan secara tempatan dengan berkesan. Selain itu, kajian ini menyediakan satu kerangka yang boleh diulang dan dinilai secara sistematik untuk pemilihan, pengoptimuman, dan penilaian komponen RAG menggunakan kaedah penilaian automatik.

Kata kunci: Retrieval-Augmented Generation (RAG), Soal Jawab Akademik, Model Bahasa Berskala Besar (LLM), Pelaksanaan Tempatan, Penilaian Automatik.

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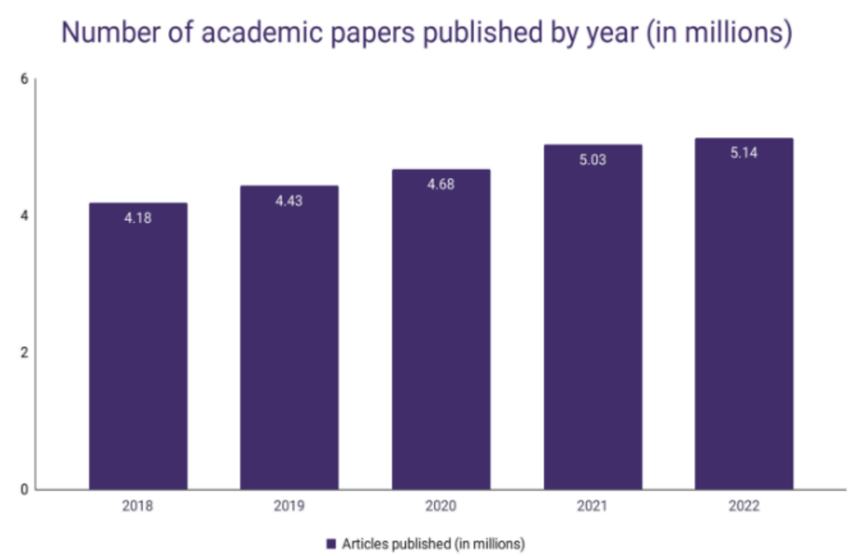
CHAPTER 1: INTRODUCTION

1.1 Research Background & Motivation

The rapid growth of academic publishing has outpaced the capabilities of traditional information retrieval systems, creating a landscape where the sheer volume of data obscures its utility. Recent estimates indicate that over 5.14 million academic articles are now published annually, reflecting a growth rate of approximately 5.6% per year (Hanson et al., 2024; Curcic, 2023).



Curcic, D. (2023). Number of Academic Papers Published Per Year – WordsRated. Wordsrated. https://wordsrated.com/number-of-academic-papers-published-per-year/

Figure 1.1: Annual Number of Academic Papers Published Worldwide (2018–2022)

While this growth accelerates scientific discovery, it also increases the difficulty of efficiently locating relevant information from a large number of articles. Conventional academic search systems often rely on lexical matching techniques to retrieve relevant documents, which are commonly referred to as **sparse retrieval** in modern-day. However, this technique struggles to capture semantic relationships between queries and documents, particularly when different terminology is used to describe the same concept (Li et al., 2017). This vocabulary mismatch problem reduces retrieval effectiveness in specialized research domains and requires researchers to manually examine numerous search results to identify relevant information (Hassoun et al., 2023; Zhu et al., 2024). To overcome this, recent information retrieval systems have increasingly adopted **dense retrieval** methods, which represent both queries and documents as dense vector embeddings. It computes the semantic similarity of the query and document, making it more effective to identify the different words with similar concepts (Karpukhin et al., 2020).

In parallel, Large Language Models (LLMs) have demonstrated remarkable capabilities in human language understanding, summarization, and question answering. However, the question answering using LLMs alone is likely to produce hallucinated or outdated information when responding to domain-specific questions (Lewis et al., 2021). This is because their response are generated based on the internal knowledge, which is usually outdated and not precise enough. Such limitation causes the question answering on the LLM alone to be unreliable in academic environments.

To solve these problems, **Retrieval-Augmented Generation (RAG)** has emerged as an effective approach by integrating document retrieval with language generation. Specifically, the RAG retrieves the relevant documents from the external knowledge base using the retriever and generates a relevant response based on the retrieved context. Instead of relying solely on the internal knowledge of the LLM during the pretraining, this approach improves answer groundedness and reduces hallucinations (Lewis et al., 2021; Gao et al., 2023).

1.2 Problem Statement

Retrieval-Augmented Generation (RAG) has been proposed for knowledge-intensive tasks to solve the problem of LLMs' difficulties in locating and generating relevant information. Its application to academic articles remains underexplored, while the existing studies focus on other domains such as healthcare and software engineering (Cremaschi et al., 2025; Hostnik et al., 2025). These existing studies are difficult to integrate with the academic document retrieval function.

The main reason is that the performance of an RAG system is highly domain-dependent. The selection of both the retrieval model and the generator models results in varying retrieval capabilities, further affecting the final response produced by LLMs. In addition, current RAG frameworks rely on cloud-based LLMs, which introduce barriers for students and researchers. The limitations include high research cost, limited customization, and privacy risks (Kwon et al., 2025; Salim et al., 2025). The privacy and copyright problem is especially important when dealing with confidential or unpublished manuscripts.

Another challenge lies in the evaluation of RAG systems. Many existing works continue to rely on human evaluation, which is subjective, time-consuming, and difficult to reproduce (Nishimura et al., 2024). The lack of standardized automatic evaluation makes it challenging to objectively compare different RAG configurations and optimize system performance.

Therefore, there is a need for a locally deployed Retrieval-Augmented Generation pipeline for academic question answering. The pipeline needs to be constructed by systematically evaluating different retrieval and generator models using automatic evaluation metrics, followed by optimization of the selected components to enhance overall pipeline performance for the academic document question-answering task.

1.3 Research Question (RQ)

RQ1: What is the technical framework required to develop a local RAG system for understanding and answering questions in the academic articles domain?

RQ2: Which retrieval model and generator model achieved the best performance for local academic question answering?

RQ3: How to determine the RAG pipeline components without relying on human evaluation?

RQ4: How does the performance of the optimized RAG pipeline compare with the baseline pipeline based on the RAGAS evaluation approach?

1.4 Research Objective (RO)

RO1: To design and develop a local Retrieval-Augmented Generation (RAG) pipeline capable of retrieving and generating answers from academic articles using local language models and vector databases.

RO2: To identify the optimal retrieval model and generator model for the proposed RAG system using automatic evaluation metrics.

RO3: To evaluate and compare the performance of the baseline and optimized RAG pipelines using answer relevancy, faithfulness, context precision, and context recall metrics.

1.5 Research Scope

This research is limited to a certain scope to ensure the problem is examined within a controlled and manageable environment. The scope of this study is as follows:

1. The proposed Retrieval-Augmented Generation (RAG) system is designed specifically for academic question answering using **English-language academic articles**. Therefore, only English-language models are considered in the study.
2. This research focuses only on **small open-weight language models (below 8 billion parameters)** to ensure their feasibility to be implemented locally.

1.6 Significance of The Study

This study aims to contribute to **Sustainable Development Goal 4, Quality Education**, by developing a Retrieval-Augmented Generation (RAG) system for academic question answering. The project provides a systematic framework for designing, optimizing, and evaluating a locally deployed RAG pipeline.

Specifically, the research demonstrates the effectiveness of combining lightweight open-source embedding models and language models with automatic evaluation metrics to identify suitable local pipeline components without relying on subjective human evaluation.

The proposed system has the potential to support teaching, learning, and research activities in higher education by facilitating more efficient literature exploration and reducing the effort required to locate relevant information.

CHAPTER 2: LITERATURE REVIEW

2.1 Type of Retrieval-Augmented Framework

Since the introduction of RAG, researchers have proposed multiple architectural variants to enhance the overall performance. These variants result in different retrieval strategies, components of the system, and the architecture's computational requirements. Table 2.1 summarizes the major types of RAG frameworks reported in recent literature.

Table 2.1: Major Retrieval-Augmented Framework Architectures

Type	Key Features	Relevant Research
Naïve RAG	Keyword-based retrieval (TF-IDF, BM25)	Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2021). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.
Advanced RAG	Dense retrieval (DPR), neural re-ranking, multi-hop retrieval	Chen, L.-C., Pardeshi, M. S., Liao, Y.-X., & Pai, K.-C. (2025). Application of retrieval-augmented generation for interactive industrial knowledge management via a large language model. Computer Standards & Interfaces, 94, 103995. https://doi.org/10.1016/j.csi.2025.103995
Modular RAG	Hybrid retrieval (sparse + dense), tool/API integration, composable pipelines	Hanmante, S., Patil, S., & Shahade, A. K. (2026). A multi module a.i. system for intelligent health insurance support using retrieval augmented generation. Scientific Reports, 16(1). https://doi.org/10.1038/s41598-025-31038-6

Graph RAG	Graph-based structures, relational reasoning, hierarchical knowledge	Lu, H., Zheng, D., & Yang, L. (2026). Mutual information-based retrieval-augmented generation for domain question answering. Knowledge and Information Systems, 68(1). https://doi.org/10.1007/s10115-025-02624-x
Agentic RAG	Autonomous agents, dynamic decision-making, iterative refinement, workflow optimization	Li, M., Zhou, Q., Li, W., Qu, T., Yang, M., & Jiang, P. (2026). A4PS: Agentic AI-assisted advanced planning and scheduling with large language models for smart manufacturing. Journal of Manufacturing Systems, 85, 207–226. https://doi.org/https://doi.org/10.1016/j.jmsy.2026.01.003

As shown in Table 2.1, **Naïve RAG** represents the earliest implementation of the RAG framework, which utilizes the sparse retrieval algorithms such as BM25 and TF-IDF to identify relevant documents. Although sparse retrieval remains effective in current applications, they rely heavily on matching the exact keywords in searching documents, which still limits its usability in higher-level tasks.

To address these limitations, **Advanced RAG** enhanced semantic search by introducing the implementation of the dense retriever using embedding models. In more recent architectures, researchers further extend the RAG frameworks with different strategies. **Modular RAG** introduced the hybrid retrieval strategies and API tools calling. **Graph RAG is** working on explicit knowledge representation using graph structures, while **Agentic RAG** proposed autonomous decision-making capabilities by the AI agents.

Despite the improvement in performance and the execution of more complex tasks, these advanced architectures also demand considerably more computational resources. Compared with conventional RAG pipelines, their reliance on graph databases or agent orchestration frameworks makes these frameworks difficult to implement locally.

Considering the objective of this study, the proposed pipeline of this study adopts an architecture that **incorporates the semantic retrieval capabilities of Advanced RAG together with the hybrid retrieval strategy of Modular RAG**. This pipeline does not contain the interchangeable modules and external tool integration, maintaining relatively low implementation complexity for local deployment.

2.2 RAG Applications Across Domains

The RAG has been successfully applied across multiple domains, especially the practical approach for improving domain-specific question answering. This is because it demonstrates the capability of generating grounded answers with retrieved documents. Table 2.2 presents several recent studies on RAG implementation across numerous domains.

Table 2.2: Summary of RAG Applications Across Domains

Reference	**Domain**	**Research Objective**	**Key Implementation**
Park et al., (2025)	**Legal**	Develop a standardized legal RAG evaluation framework	LRAGE benchmark for legal RAG eval
Hostnik et al., (2025)	**Software Dev**	Improve code suggestion accuracy with retrieval	Local RAG for code completion
Cheng et al., (2025)	**Healthcare-like**	Analyze RAG effectiveness across domains	Survey on knowledge RAG apps

Shi et al., (2025)	**Software Dev**	Enhance LLM security via external knowledge retrieval	RESCUE: hybrid RAG for secure code gen

The studies summarized in Table 2.2 demonstrate that RAG consistently improves task performance across different domains. For instance, Park et al. (2025) introduced an evaluation framework to integrate RAG on legal document analysis, whereas Hostnik et al. (2025) applied RAG to software development to improve code completion. In addition, Cheng et al. (2025) highlighted the applicability of RAG on healthcare-related tasks, while Shi et al. (2025) demonstrated that hybrid retrieval improves the reliability of LLM-assisted code generation. While RAG has demonstrated promising performance in domains such as legal services, healthcare, and software engineering, its application to academic literature remains relatively limited.

2.3 Cloud-Based RAG Models

Most of the RAG implementations rely on employing cloud-based language models due to their superior reasoning capabilities and ease of integration through API, while this also introduces several limitations. Table 2.3 summarizes several representative cloud-based RAG implementations.

Table 2.3: Comparison of Cloud-Based RAG Models and Their Constraints

Reference	Framework/Model	Cloud Dependency	Research Objective	Key Drawbacks
Lewis et al., (2021)	BART/DPR RAG	Cloud TPUs for training	Enable knowledge-intensive NLP via retrieval	**Cutoff dates, API reliance**

Brehme et al., (2025)	Cloud-hosted RAG	External APIs for eval	Assess LLM judging reliability for RAG	**Trust issues, privacy leaks**
Park et al., (2025)	GPT/cloud retrievers	Cloud embeddings	Benchmark retrieval in legal contexts	**Legal data sovereignty risks**
Kersting et al., (2026)	Azure GPT-4 RAG	Azure OpenAI APIs	Compare cloud vs local RAG for PLC code	**High costs for industrial scale**

As shown in Table 2.3, the reviewed studies applied to cloud-based RAG systems share similar characteristics, although they target different application domains. While these works demonstrate a strong performance in the RAG framework, they share a common dependency on third-party cloud providers. This dependency simplifies deployment but reduces user control over model configuration and data processing.

The dependency introduces several practical limitations. These include recurring operational costs, limited model customization, potential privacy concerns when processing sensitive documents, and reliance on stable internet connectivity. These limitations motivate the development of locally deployed RAG systems that provide greater privacy, flexibility, and cost efficiency without relying on external cloud services.

2.4 Human Evaluation on RAG Models

Evaluating the effectiveness of the RAG pipeline is as important as designing the pipeline itself. Although automatic evaluation has gained increased attention in recent years, many of the existing studies still rely on human assessment to determine the quality of the generated response. Table 2.4 presents the recent studies using human interference as the gold standard.

Table 2.4: Human Evaluation Methods for RAG Pipeline Assessment

Reference	Evaluation Method	Research Objective	Research Result	Issues with Human Eval
Mangold et al., (2025)	Human ratings/Likert	**Validate human-centered RAG metrics**	Human prefs align 65% with auto-scores	**Scalability, bias issues**
Brehme et al, (2025)	Human gold labels	**Compare human vs LLM evaluation**	LLM judges match humans at 82% macro-F1	**Inter-annotator disagreement**
Lee et al, (2025)	Human QA annotation	**Create a difficulty matrix for RAG QA**	Identified 3 difficulty tiers via annotation	**Labor-intensive difficulty labeling**
Yang et al, (2025)	Developer surveys	**Evaluate RAG utility in real dev envs**	lexical + semantic retrieval boosts pass@1 (first-attempt pass rate) by 15%	**Survey bias, small sample**

The reviewed studies demonstrate that human evaluation remains a common approach across many different applications. For instance, Lee et al (2025) used manual annotation to classify question difficulty. Some studies have noticed the problem and performed research on it. Brehme et al. (2025) compared human judgments with LLM-based evaluation, and Mangold et al. (2025) employed Likert-scale ratings to assess human-centered evaluation metrics.

The dependence on human evaluation has presented several limitations. It introduces the evaluator bias and inter-annotator disagreement throughout the research. Evaluating the RAG pipeline becomes less scalable and difficult to reproduce similar results consistently. Therefore, this study adopts automatic evaluation metrics such as the RAGAS evaluation framework, aiming to provide a more reproducible assessment of the proposed RAG pipeline.

CHAPTER 3: METHODOLOGY

3.1 Dataset Description

For this research, we utilize the arXiv Dataset, a large collection of scholarly articles originally published by Cornell University and made publicly available via the Kaggle website. The arXiv repository provides free access to the latest articles across various scientific fields, which enables us with a quality data source to train and evaluate our RAG pipeline effectively. The selected dataset from Kaggle consists of **136238** rows and **10** attributes.

Table 3.1: Attributes of the selected dataset

Column Name	**Description**
id	Unique identifier for each paper from arXiv.
title	Full title of the research paper.
summary	Abstract or summary text of the paper.
summary_word_count	Number of words in the summary text.
category	Subject area(s) assigned by arXiv (e.g., physics, cs.AI).
category code	Coded version of the subject category.
published_date	Date when the paper was first published on arXiv.
updated_date	The most recent date the paper's record was updated.
authors	List of all authors of the paper.
first_author	First author listed on the paper.

The dataset was selected because it provides both article metadata and a dedicated **summary** field, making it particularly suitable for Retrieval-Augmented Generation (RAG) applications. The summary content serves as the primary knowledge source for document retrieval and response generation, while also providing a reference text for evaluating the semantic consistency of generated responses through Natural Language Inference (NLI)-based assessment.

In addition, the dataset supports the construction of evaluation datasets with known source documents. By generating questions from individual articles and preserving the associated document identifiers, retrieval performance can be objectively measured using Recall@k, while the article summaries provide reference texts for semantic consistency evaluation.

3.2 Sampling and Benchmark Question Design

The following section describes the data sampling strategy and benchmark questions set design for the experimental evaluation purpose.

3.2.1 Dataset Sampling Strategy

To construct the evaluation datasets, five research fields were selected from the arXiv dataset: **Machine Learning (ML), Statistics (Stats), Computer Vision (CV), Natural Language Processing (NLP), and Artificial Intelligence (AI).** For each research field, four publication year groups were created to represent different temporal scopes of the articles. A **stratified random sampling** approach is used during the process. The dataset was first divided according to research field and publication year group. Samples were then randomly drawn from each cluster to ensure a balanced representation across domains and temporal periods. As a result, **20 datasets** (5 research fields × 4 year groups) were created and ready to store into the vector database.

Table 3.2.1: Publication Year Groups and Maximum Sample Sizes per Research Field

Year Group	**Description**	**Maximum Samples per Research Field**
2024	Current	100
2023–2024	Recent	200
2020–2024	Modern	500
2015–2024	Decade	1,000

This dataset design enables the evaluation of the proposed RAG pipeline across different research domains while also examining the impact of collection size and temporal coverage on retrieval and generation performance.

3.2.2 Benchmark Question Construction

Based on the dataset created in the previous stage, questions are constructed from the sampled articles and linked to their corresponding documentIDs and summaries. A benchmark question set comprised of **20** questions was constructed for each dataset. With 20 datasets in total, this results in **400 unique questions** for the following evaluation. stage. The generated question categories access the pipeline from simple to more complex tasks, ranging from factual retrieval to multi-document reasoning and refusal behavior.

The questions are distributed across the following categories:

- **Direct Fact Retrieval and Definition:** The most direct question type that requires the pipeline to retrieve specific factual information from a source document.

- **Summarization and Purpose Identification:** This type of question asks the pipeline to perform summarization on a single article.
- **Comparative and Contrastive:** Questions that require comparison across **multiple source documents**. The questions test the pipeline's ability to understand and distinguish between information from different sources.
- **Synthesis and Multi-Hop Reasoning:** Similar to the previous category, this requires integrating information from **more than one document**. These questions assess the pipeline's ability to combine evidence from different articles and produce a final answer.
- **Negative / Unanswerable:** Questions intentionally designed to have **no answer** within the provided corpus, used to assess the pipeline’s ability to identify insufficient information and avoid generating unsupported responses.

3.3 Proposed Pipeline

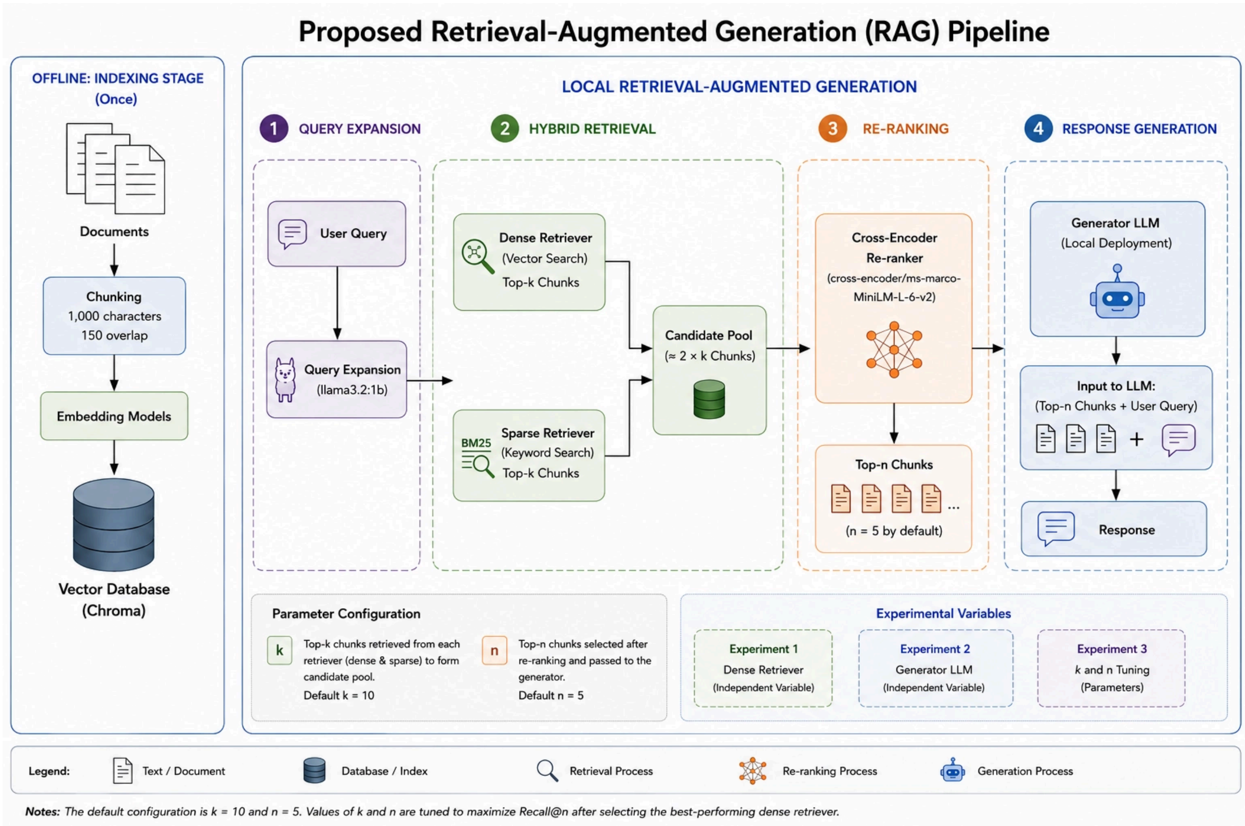


Figure 3.3: Architecture of the Proposed Local RAG Pipeline

Figure 3.3 illustrates the overall architecture of the proposed Retrieval-Augmented Generation (RAG) pipeline. The system consists of four main stages: query expansion, hybrid retrieval, re-ranking, and response generation.

The parameter k refers to the number of top-ranked text chunks retrieved from the dense and sparse retrievers, forming the candidate pool for re-ranking. A larger k increases retrieval coverage but may introduce more irrelevant chunks.

The parameter n represents the number of top-ranked chunks selected after cross-encoder re-ranking and provided to the generator model. Larger n values generally improve Recall@n by retaining more potentially relevant information, while smaller values reduce context size but may exclude useful evidence.

In this study, both k and n are treated as experimental parameters. The default configuration is $k = 10$ and $n = 5$; however, their optimal values are determined empirically after selecting the best-performing dense retriever. Since the primary objective is to maximize retrieval coverage, parameter tuning prioritizes **Recall@n** over precision.

3.3.1 Query Expansion

The pipeline starts with a query sent to the system. To improve retrieval effectiveness, the original query is first expanded into a richer representation containing additional context and related terms. The **llama3.2:1b** was selected as the query expansion model and fixed across all experiments to ensure a consistent retrieval pipeline and prevent query reformulation from influencing the comparative evaluation. The 1B-parameter version was selected because query expansion is a relatively lightweight task that does not require extensive reasoning ability from larger models, while also improving the inference speed. Lastly, the expanded query is subsequently passed to the hybrid retrieval module.

3.3.2 Hybrid Retrieval

During the retrieval stage, two complementary retrieval approaches are executed in parallel. The first is **dense vector search**, which retrieves semantically relevant text chunks from the **Chroma vector database** using embedding-based similarity matching. The embedding model of dense retriever is treated as the **independent variable** throughout the experiments, allowing its contribution to retrieval effectiveness to be evaluated independently of other pipeline components.

The second retrieval approach is **BM25 keyword search**, which retrieves text chunks based on lexical relevance. BM25 was selected as the sparse retriever due to its strong performance and being widely used in retrieval systems. Recent studies employ BM25 as a baseline sparse retriever in hybrid retrieval architectures, further demonstrating its robustness and effectiveness (Akarsu et al., 2026).

The retrieved results from both the dense vector retriever and the sparse BM25 retriever are combined to form a candidate pool of potentially relevant documents. During this process, duplicate document chunks retrieved by both methods are removed, ensuring that each document chunk appears only once in the final candidate pool. The remaining unique chunks are then ranked using the **Reciprocal Rank Fusion (RRF)** algorithm, which combines the ranking positions assigned by both retrievers. This approach boosts documents that are ranked highly by both semantic and keyword-based retrieval methods, resulting in a consolidated candidate pool that leverages the complementary strengths of dense and sparse retrieval while reducing redundancy.

3.3.3 Cross-Encoder Re-Ranking

To improve the quality of the retrieved context, a **cross-encoder re-ranker** is applied to the candidate pool generated by the dense and BM25 retrieval. The re-ranker evaluates the relevance of each query–document pair and assigns a relevance score. The candidate chunks are then reordered according to these scores, and the top **five** ranked chunks are selected as the final retrieval context for response generation.

For the re-ranking stage, the **cross-encoder/ms-marco-MiniLM-L-6-v2** model was adopted. Cross-encoders are widely used in two-stage retrieval pipelines because they jointly encode the query and document, enabling more accurate relevance estimation than independent vector similarity matching. The selected model was trained on the MS MARCO passage ranking dataset and has demonstrated strong reranking performance while maintaining relatively low computational requirements (Wang et al., 2025).

3.3.4 Response Generation

The retrieved documents are provided to the locally deployed generator together with the user query to produce the final response. Similar to the dense retriever, the generator model is treated as an independent variable throughout the second experiment, allowing different local LLMs to be evaluated under the same retrieval conditions. Furthermore, an additional fine-tuning stage will be performed to further improve its performance for academic question-answering tasks.

To ensure consistent response generation across all experiments, a fixed prompt template was employed throughout the study. The prompt instructs the model to focus only on relevant retrieved documents to avoid generating information that is not supported by the provided context. In addition, the generator needs to return **"No Answer"** when the retrieved documents do not contain sufficient evidence to answer the user's question. The retrieved document and user query will be added to the predefined prompt template in every question-answering section.

3.4 Experiment Setup

This section presents the experimental setup to evaluate the performance of the Retrieval-Augmented Generation (RAG) pipeline.

3.4.1 Experiment 1: Dense Retriever Selection

To determine the most accurate dense retriever for the RAG pipeline, a comparative evaluation was conducted among four different open-source embedding models.

Table 3.4.1: Candidate Embedding Models

Model	**Developer**	**Embedding Dimension**
all-minilm:22m	Microsoft	384
granite-embedding:30m	IBM	384
nomic-embed-text:v1.5	Nomic AI	768
qwen3-embedding:0.6b	Alibaba	1024

The selected embedding models were chosen to represent a range of model sizes, architectures, and embedding dimensions commonly used in retrieval tasks. Since the proposed RAG system is intended for local deployment, models that achieve strong retrieval performance while maintaining lower computational requirements are preferred. Therefore, the evaluation considers both retrieval effectiveness and model size, with priority given to lightweight models if comparable performance can be achieved.

The evaluation was conducted using a structured testing pipeline. For each embedding model, all queries were executed through the retrieval system under the same hardware and system configurations to ensure a fair comparison. The retrieved document for each query was recorded and compared against the corresponding ground-truth ***documentID***. Retrieval effectiveness was then measured based on the model's ability to return the correct source document.

The primary metric used to evaluate retrieval effectiveness is **Recall@n**. The evaluation metric measures the percentage of true source articles successfully retrieved in the top n chunks returned by the retriever. Formally, Recall@n is defined as:

$$Recall@n = \frac{Relevant\ Chunks\ Retrieved\ in\ Top\ n}{Total\ True\ Ground\ Truth\ Chunks}$$

Since our default value for n is 5, this experiment will calculate Recall@5 to compare retriever performance.

3.4.2 Experiment 2: Generator Model Selection

The second experiment focuses on identifying the most suitable generator model for the proposed RAG pipeline. The generator is a critical component of the pipeline because it is responsible for producing the grounded response. Unanswerable Questions categorized as *negative* or *unanswerable* were excluded from this experiment to ensure that the assessment focuses solely on answer quality.

A comparative evaluation was conducted among several open-source language models available through the **Ollama** framework. The selected models represent different model families and parameter scales, enabling a comprehensive assessment of response quality and computational efficiency. Similar to Experiment 1, models that achieve comparable performance with lower computational requirements are preferred for deployment in the proposed local RAG system.

Table 3.4.2: Candidate Generator Models

Model	Developer	Selection Rationale
llama3.2:1b	Meta	A lightweight instruction-tuned model was selected as a baseline for evaluating compact LLMs in a local RAG environment.
llama3.2:3b	Meta	A larger variant of Llama 3.2 was included to assess the impact of model scale within the same model family.
qwen2.5:3b	Alibaba	A multilingual instruction-tuned model recognized for strong language understanding and generation performance.
phi3.5:3.8b	Microsoft	A compact model known for strong reasoning and instruction-following capabilities relative to its size.
gemma2:2b	Google	A lightweight model designed to provide a balance between language performance and computational efficiency.

The evaluation was conducted using a structured testing pipeline. The dense retriever selected from Experiment 1 was fixed throughout the evaluation, and the same retrieval configuration was maintained across all generator models. For each query, the system generated a response based on the retrieved context along with the user query. Finally, the generated response was recorded for subsequent evaluation.

3.4.2.1 Semantic Consistency Evaluation (NLI Evaluation)

This semantic consistency evaluation using Natural Language Inference (NLI) models was inspired by the methodology proposed by Honovich et al. (2022). The authors utilize the NLI models to assess whether a generated text is semantically supported. This is strongly aligned with one of our research objectives, which evaluates the pipeline performance without using human interference. Following this approach, the source article summary was treated as the premise, while the generated response was treated as the hypothesis.

To further improve the robustness of the assessment, this evaluation employed three different pre-trained NLI models with different architectures. The final evaluation outcome was determined by aggregating the labels produced by the three models. Using multiple NLI models helps reduce potential bias from any single model and provides a more reliable assessment of semantic consistency.

- **facebook/bart-large-mnli -** A widely adopted NLI benchmark model included as a strong baseline for entailment classification and semantic consistency assessment.
- **cross-encoder/nli-deberta-v3-large -** A large cross-encoder NLI model selected for fine-grained assessment of semantic consistency between generated responses and source evidence.
- **MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli -** A DeBERTa-v3 model trained on multiple NLI datasets, selected for its improved robustness and generalization across diverse inference scenarios.

Before NLI evaluation, generated responses were split into individual sentences. Each sentence was then evaluated independently and labeled as entailment, neutral, or contradiction based on the semantic relationship with the corresponding source article summary. This sentence-level evaluation provides a more precise assessment than evaluating the whole response because it reduces the risk of the unsupported statement being overlooked. Two complementary metrics were used:

- **Average Entailment Score:** Measures the degree to which a generated response is semantically supported by the corresponding source article summary. The score was computed as the proportion of sentence-level classifications labeled as *entailment* within the generated response. Higher scores indicate stronger semantic alignment between the generated response and the source evidence.

- **Majority-Vote Classification:** For each NLI model, a response-level label is determined through majority voting across its sentence-level classifications. Subsequently, the final response-level label is obtained through a second majority voting process across the three NLI models, providing a more robust assessment of semantic consistency.

3.4.2.2 Refusal Capability Evaluation

This experiment assesses the model's performance when dealing with **Unanswerable Questions**. These question categories were specifically designed to evaluate the ability of generator models to identify when the provided information is insufficient to answer a question and to appropriately refuse to generate an unsupported response.

The model refusal capability was measured using the **Refusal Rate**, defined as the percentage of such questions for which the model correctly refrained from providing an unsupported answer. To improve the robustness of the evaluation, refusal decisions were assessed using two separate approaches. The results from both methods were analyzed together to provide a more reliable assessment of the model's refusal behaviour.

- **Keyword-Based Refusal Detection:** Identifies predefined refusal phrases within the generated response.
- **NLI-Based Refusal Detection:** Follows the same NLI evaluation process as described in the previous section. However, this method evaluates whether the generated response

indicates that the availability information is not sufficient to answer the question, instead of comparing with the source document.

3.4.3 Experiment 3: Retriever–Generator Combination Analysis and Optimization

The final experiment focuses on optimizing both the retriever and generator components of the proposed RAG pipeline. At the end of the section, we evaluate the RAG pipeline as an integrated system by comparing the performance of the optimized pipeline against the baseline configuration using the RAGAS evaluation approach.

3.4.3.1 Retriever hyperparameter tuning

The first experiment helps to select the appropriate embedding model as the retriever. This part of the study optimizes it by **identifying the optimal values of *k* and *n*.** A grid search approach was employed for the hyperparameter tuning, with $k \in \{5, 10, 15\}$ and n ranging from 1 to 10, subject to the constraint that $n \leq k$ to ensure that the re-ranking stage always operates on a valid candidate set.

Similar to Experiment 1, **Recall@n** was used as the primary evaluation metric. Instead of simply maximizing recall, the optimal configuration was selected based on the **point of diminishing returns,** corresponding to the elbow point in the Recall@n curve where marginal gains begin to stabilize.

3.4.3.2 Fine-Tuning the Generator Model

In addition to retriever optimization, the selected generator was also further enhanced through the fine-tuning process. The procedure consisting of constructing a dedicated dataset, training the model using QLoRA, and deploying the trained model on Ollama, details are listed below:

1. **NLI-Based Data Filtering** – High-quality responses were selected from previous NLI evaluation results.
2. **Deduplication and Stratified Sampling** – Removing the duplicated questions by only keeping the highest-scoring response for each question. Then apply the Stratified sampling to ensure balanced coverage across different research domains and question categories.
3. **Synthetic Data Generation** – Inferencing a larger Language Model (**llama3:8b**) to create different response variations for Qwen2.5 to learn with.
4. **Dataset Validation and Preparation** – Merging the real and synthetic datasets and performing additional NLI filtering and deduplication again before converting into the **ChatML supervised fine-tuning (SFT)** format required by Qwen2.5.
5. **QLoRA Fine-Tuning** – The SFT format of the dataset is used to fine-tune **Qwen2.5-3B** using Quantized Low-Rank Adaptation (**QLoRA**). This method only updates the adapter parameters while keeping the quantized base model frozen.
6. **Model Merging and Deployment** – The trained LoRA adapters were merged with the base model and converted to GGUF format for efficient deployment via Ollama. As a result, the **optimized Qwen2.5-3B** model is ready to use.

3.4.3.3 RAGAS evaluation framework

The effectiveness of the optimized RAG pipeline was evaluated using the **RAGAS evaluation framework.** This approach measures both retrieval and generation performance through four metrics, which are **Answer Relevancy**, **Faithfulness**, **Context Recall**, and **Context Precision**.

The **Answer Relevancy** evaluates how well the generated response addresses the user's question. The **cross-encoder/ms-marco-MiniLM-L-6-v2** model is used to measure the evaluation metric. It processes the query and response to produce a relevance score that ranges from 0 to 1, where a higher score indicates a stronger semantic correlation between the query and the response.

The Faithfulness uses a **cross-encoder/nli-deberta-v3-base model** to evaluate whether the retrieved context supports the generated response. The procedure is similar to Experiment 2, which is also assessed through a sentence-level NLI approach.

For retrieval evaluation, **Context Recall** and **Context Precision** were calculated using ID-based matching. Calculating the metrics directly against the ground-truth documentIDs instead of calling LLM models as a judgment. This approach provides the most direct assessment of retrieval performance without using additional evaluation models.

To provide a clearer overview of system performance, we calculate the mean of the Answer Relevancy and Faithfulness to produce the **Generator Metrics Average.** On the other hand, the Context Recall and Context Precision were also combined into a **Retriever Metrics Average.** Finally, the overall **RAGAS Index** was calculated as the average of all four metrics.

CHAPTER 4: ANALYSIS AND RESULTS

4.1 Experimental Results for Dense Retriever Selection

4.1.1 Overall Retrieval Performance

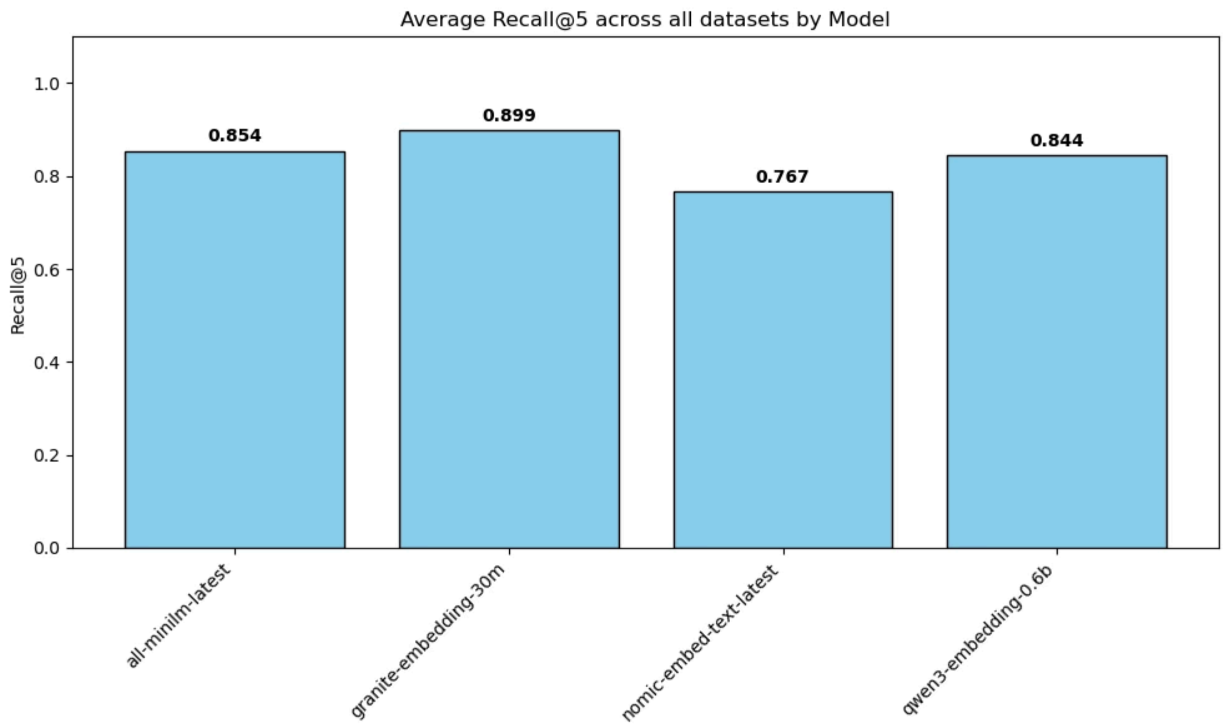


Figure 4.1.1: Average Recall@5 Performance by Dense Retriever Embedding Model

As illustrated in Figure 4.1.1, **granite-embedding-30m** achieved the highest average Recall@5 at **89.90%**. This was followed by **all-minilm-latest** and **qwen3-embedding-0.6b**, which both achieved a similar performance with Recall@5 of 85.40% and 84.43%. While **nomic-embed-text-latest** scored 76.67% on the evaluation metrics, making it the worst performance in academic document retrieval in all four models. To summarize, all evaluated models achieved Recall@5 scores above 75%, indicating that each embedding model was capable of retrieving a sizable proportion of the ground-truth documents.

4.1.2 Performance Breakdown by Dataset Year Group

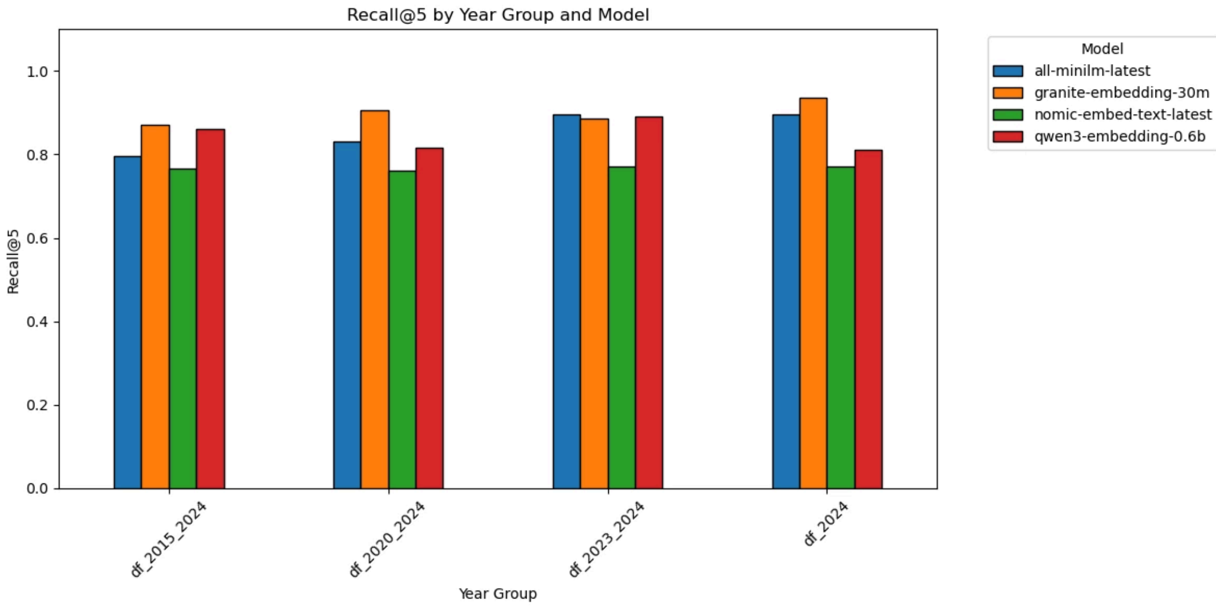


Figure 4.1.2: Recall@5 Performance by Publication Year and Embedding Model

To further evaluate retrieval performance across different temporal domains, this section presents the performance of each embedding model across different year groups.

The results show that **granite-embedding-30m** consistently achieved the highest retrieval performance across all year groups, peaking at **93.60%** on the df_2024 dataset. Similarly, **all-minilm-latest** maintained stable performance and exhibited a gradual improvement as the dataset became more recent, achieving approximately **90.00%** Recall@5 on both **df_2023_2024** and **df_2024**.

Meanwhile, **qwen3-embedding-0.6b** demonstrated unstable performance across the year groups, with Recall@5 ranging from approximately **81.00%** to **89.00%**. In contrast,

nomic-embed-text-latest displays the most stable but lowest performance, maintaining Recall@5 scores between **76.00%** and **77.00%** across all datasets.

Overall, **granite-embedding-30m** remained the best performance model across all temporal groups. This further supports its selection as the dense retriever for the proposed pipeline.

4.2 Experimental Results for Generator Model Selection

4.2.1 Semantic Consistency Evaluation Results

This part of the study evaluates the semantic consistency of the generated output using the NLI approach as described in Chapter 3. Only answerable questions were included in this experiment. The model performance is measured using two methods, which are Average NLI Entailment Score and Majority-Vote NLI label distribution.

4.2.1.1 Average NLI Entailment Score

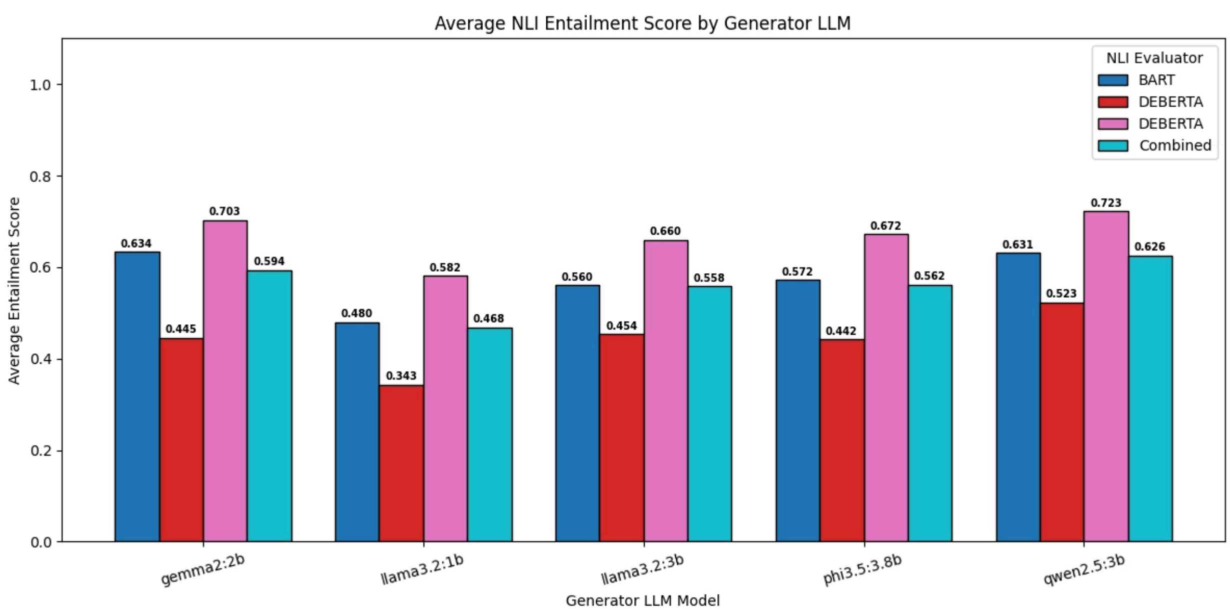


Figure 4.2.1.1: Average Sentence-Level NLI Entailment Scores by Generator LLM

Figure 4.2.1.1 illustrates the entailment score evaluated using different NLI evaluators, along with their combined average. Among all candidate generators, **qwen2.5:3b** achieved the highest combined score with 0.626, indicating the most consistent between the generated contents and source article summaries. In contrast, the **llama3.2:1b,** which served as the baseline model, recorded the lowest score of 0.468. This demonstrates the performance difference between the strongest and the weakest model in the selected generator candidates.

A similar performance trend was observed across the **phi3.5:3.8b, llama3.2:3b, and gemma2:2b**. Although these models exhibit competitive performance, their combined entailment scores are consistently below those of the **qwen2.5:3b**. In conclusion to this part, the **qwen2.5:3b** frequently outputs the response that is supported by source articles as compared to other generators.

4.2.1.2 Majority-Vote NLI label distribution.

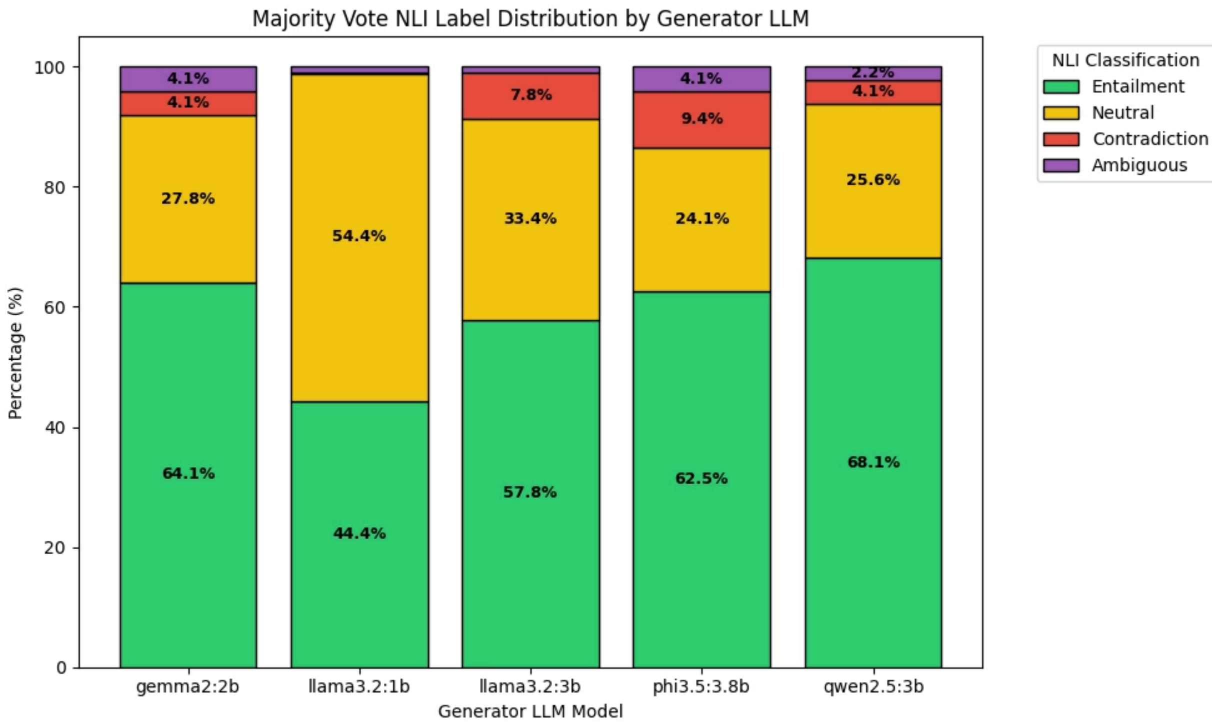


Figure 4.2.1.2: Response-Level NLI Classification Distribution by Generator LLM

Figure 4.2.1.2 presents the distribution of the response level majority vote classification of each generator model. The results are consistent with the previous experiment, the **qwen2.5:3b** achieved the highest proportion of entailment classifications, with **68.1%** of the responses being majority vote as entailment from 3 NLI models. This result is followed by **gemma2:2b (64.1%)** and **phi3.5:3.8b (62.5%)**. While the llama-based model is demonstrated as underperforming, with **llama3.2:3b(57.8%)** and **llama3.2:1b(44.4%)**.

Besides the entailment label, the contradiction rates remained relatively low across all models, ranging from **4.1%** to **9.4%**. The llama3.2:1b, despite having the lowest contradiction percentage, showed the highest neutral classification rates **(54.4%)**. This result suggests that its response lacks sufficient evidence for the definitive entailment label, while also not completely contradicting the source document, indicating unclear outputs by the generator.

4.2.2 Refusal Capability Evaluation Results

This experiment assesses the candidate generators on handling Unanswerable questions.

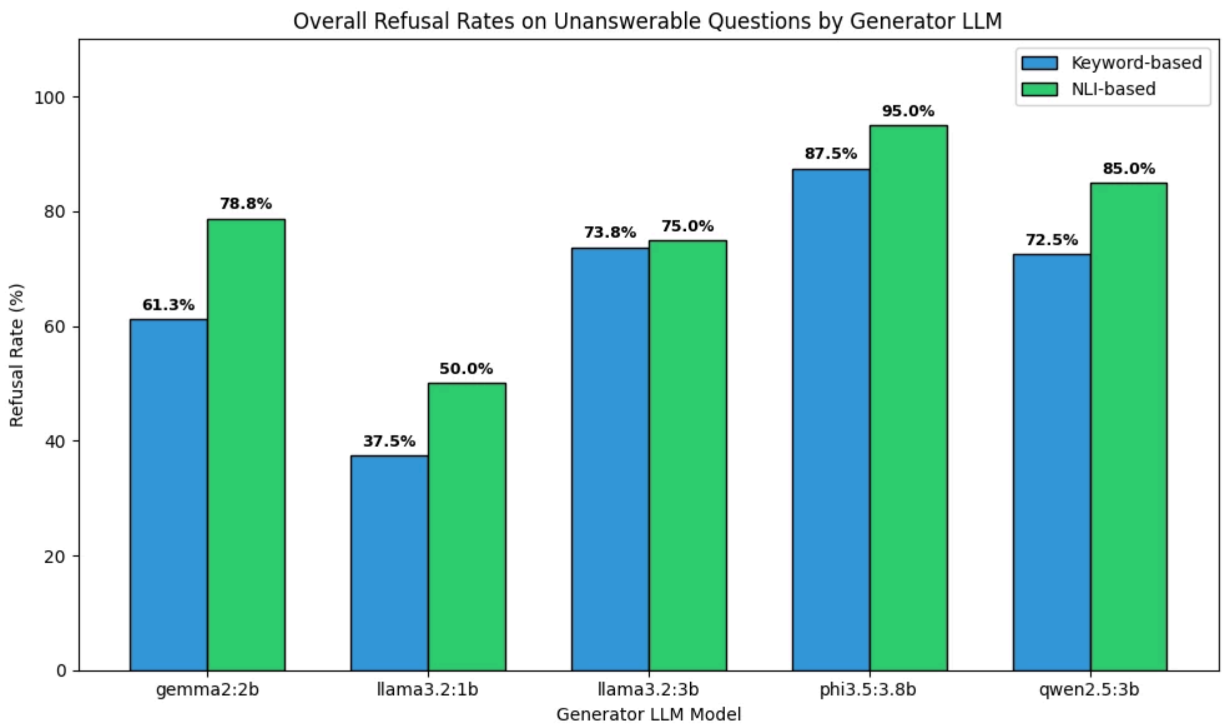


Figure 4.2.2: Keyword-Based and NLI-Based Refusal Rates for Candidate Generator LLMs

Figure 4.2.2 presents the refusal rates achieved by candidate generators evaluated using a keyword-based and an NLI-based approach. Among all candidates, the **phi3.5:3.8b** showcases the best refusal performance, scoring a refusal rate of up to 95% in NLI-based evaluation. In contrast, **llama3.2:1b** continues to record the lowest scoring, with a refusal rate of only 37.5% and 50.0% in both approaches. The **gemma2:2b** model illustrates the high difference in the two scoring methods, 61.3% in keyword-based and 78.8% in NLI-based. This indicates that the model did not use the predefined keywords to refuse answering the question. Overall, except for **llama3.2:1b**, all candidate generators demonstrated high refusal rates, indicating their ability to appropriately identify the unanswerable questions and respond with refusal output.

4.2.3 Final Model Selection

Based on the results of the semantic consistency and refusal capability evaluations, **qwen2.5:3b** was selected as the generator model for the proposed RAG pipeline. The model achieved the highest Average Entailment Score (0.626) and Majority-Vote Entailment Rate (68.1%) while maintaining strong refusal performance. Therefore, **qwen2.5:3b** was selected for all subsequent experiments and the final proposed RAG system.

4.3 Retriever–Generator Combination and Optimization

4.3.1 Retriever Hyperparameter Tuning

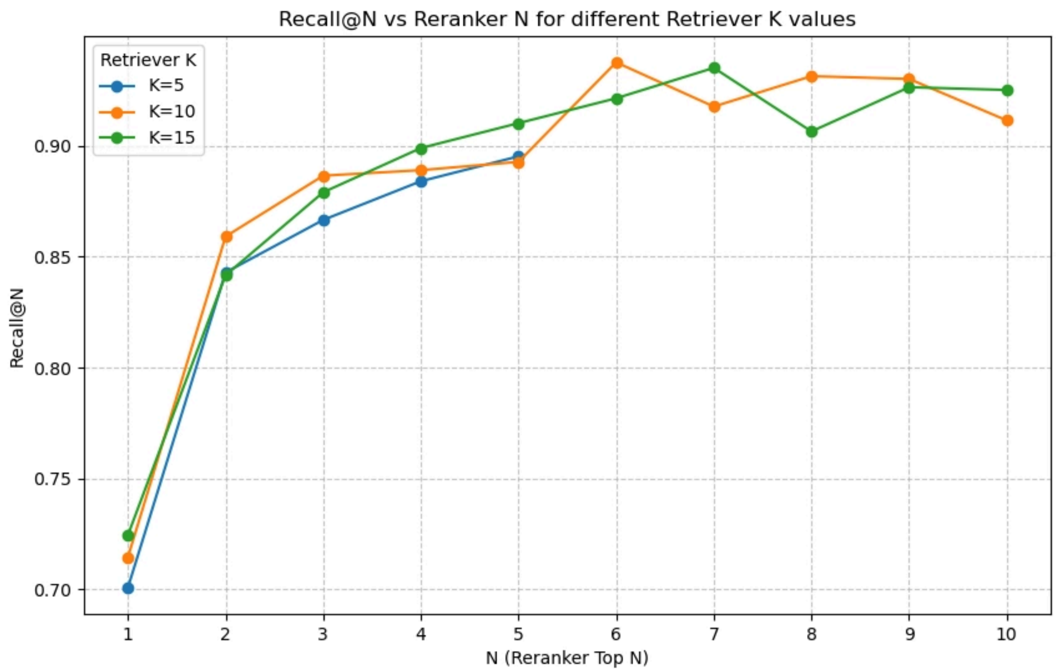


Figure 4.3.1: Hyperparameter Grid Search Results Under Different Candidate Pools

Figure 4.3.1 presents the Recall@n performance under different retriever configurations, where *k* represents the number of candidate documents retrieved by the retriever and *n* represents the number of documents retained after the re-ranking stage.

Based on the line chart, the Recall@n increased rapidly before n = 3 and became more gradual after that, indicating that the benefit gained from increasing n slowly decreased. In contrast, varying *k* had relatively little impact on Recall@n, with the performance curves for **k = 10** and **k = 15** remaining consistently close throughout the evaluation.

After review, **k = 10** and **n = 6** were selected as the optimal retriever configuration for the proposed RAG pipeline. This setting corresponds to the elbow point of the Recall@n curve and achieves the highest Recall@n among all evaluated configurations. Considering that the optimized retriever will deploy locally, increasing k beyond **10** did not provide additional retrieval benefits while also increasing the computational overhead.

4.3.2 RAGAS Evaluation Result

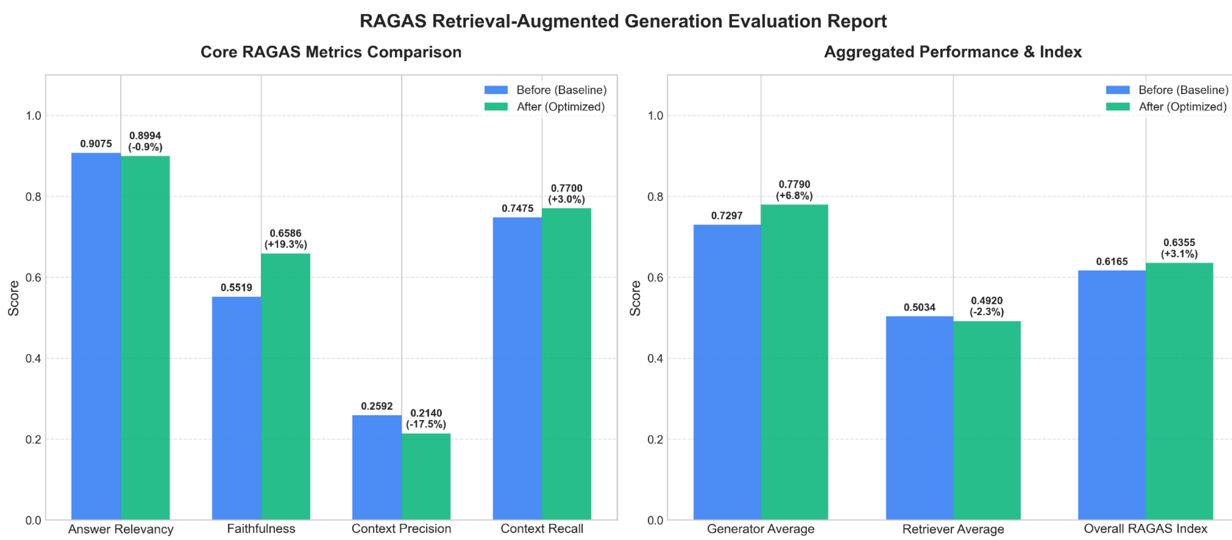


Figure 4.3.2: RAGAS Performance Before and After Pipeline Optimization

Figure 4.3.2 compares the RAG pipeline performance before and after the optimization using the RAGAS evaluation framework. Throughout the evaluation, the most significant improvement was the RAG Faithfulness. The **Faithfulness** increased from 0.5519 to 0.6586, exhibiting a 19.3% improvement. On the retriever metrics, the **Context Recall** gains a slight improvement from 0.7475 to 0.7700, demonstrating a 3% improvement.

In contrast, **Context Precision** decreases from **0.2592** to **0.2140**. This reduction was observed alongside the increase in **Context Recall**, reflecting the tradeoff during the optimization process. On the other hand, the **Answer Relevance** remains almost unchanged on the baseline and optimized pipeline. Maintaining a high score before (0.9075) and after (0.8994) the optimization. The overall RAGAS score gains a stable increase from **0.6165** to **0.6355**, suggesting an improvement in overall performance.

CHAPTER 5: DISCUSSION

5.1 Discussion on Dense Retriever Selection

The retriever evaluation results show that **granite-embedding-30m** consistently outperformed the other embedding models. By referring to Recall@5 performance, this model indicates a strong ability to retrieve relevant source documents across every year group. This ability of the model is critical for ensuring that the following generator receives accurate documents and evidence to produce faithful responses.

As mentioned previously, smaller models are generally preferred when they can achieve performance comparable to larger alternatives due to their lower computational requirements. In this study, an interesting observation is that retrieval performance was not directly correlated with model size. Despite having substantially more parameters, **qwen3-embedding-0.6b** did not outperform **granite-embedding-30m**, which is considerably smaller in size. The results indicate that the retrieval effectiveness does not solely depend on the model scale, many other factors, such as the model's training methodology and its suitability for the retrieval task, are also important criteria that need to be considered.

5.2 Discussion of Generator Model Selection

The results demonstrate that the performance differences were more apparent in the majority-vote classification results than in the average entailment scores, suggesting that categorical NLI labels provide a clearer distinction between the semantic consistency capabilities of different generator models. Furthermore, higher refusal rates observed under NLI-based evaluation indicate that some refusals were expressed implicitly, highlighting the importance of using multiple evaluation metrics to obtain a more comprehensive assessment of model performance.

Similar to the findings in the Retrieval phase, the larger model size does not necessarily translate into higher scoring on our experiments. While having the largest number of parameters among all evaluated models, **phi3.5:3.8b** only achieved the highest refusal rate but did not outperform the **qwen2.5:3b** in semantic consistency. In contrast, **gemma2:2b** demonstrated competitive performance despite its smaller size. These results suggest that model architecture and training quality may have a greater influence than parameter count alone, as reflected by **llama-based models** underperforming in the study.

Overall, **qwen2.5:3b** achieved the highest semantic consistency scores while maintaining strong refusal capability, making it the selected generator for this proposed RAG pipeline.

5.3 Discussion of Combination Analysis

The results of Experiment 3 demonstrate that optimizing both the retriever and generator components can improve the ability of the proposed RAG pipeline. The optimization process focused on improving retrieval coverage and reducing unsupported generation, which are critical requirements for academic question-answering tasks.

The increment of the re-ranking N increases the likelihood of retrieving non-relevant documents, thus causing the decrease of the **Context Precision** metric. This outcome was expected during the optimization process, as the project placed greater emphasis on **Context Recall** than on Precision. Retrieving relevant documents is more crucial to provide the generator with sufficient evidence to produce a faithful response. Furthermore, the selected candidate generator having decent context size to handle a large amount of input during response generation.

The fine-tuning of **Qwen2.5-3B** demonstrates the application of data analytics techniques in constructing a high-quality training dataset through NLI-based filtering, deduplication, stratified sampling, and synthetic data augmentation. The **QLoRA** fine-tuning approach updates only adapter parameters, reducing computational costs while preserving the capabilities of the pre-trained model. As a result, the optimized generator achieved a substantial improvement in **Faithfulness**, indicating that its responses were more consistently supported by the retrieved context. In contrast, **Answer Relevancy** remained high and stable between the baseline and optimized pipelines. This suggests that the baseline model was already capable of producing responses that were semantically aligned with user queries, while the fine-tuning process could not further improve on this part.

In summary, although the overall increase in the RAGAS Index appears modest, the improvements occurred in the most critical metrics for the proposed academic RAG system, namely Faithfulness and Context Recall. These metrics directly reflect the system's ability to retrieve relevant information and generate responses that are supported by evidence from academic articles.

CHAPTER 6: CONCLUSION

This study proposed and evaluated a locally deployed RAG pipeline for the academic question-answering task using an automatic evaluation method. The primary motivation is to fill the gap in the lacking RAG-supported academic search system implementation on open-weight local language model and vector database.

The first research objective was successfully achieved by the proposed pipeline, which integrates query expansion, hybrid retrieval, a reranking mechanism, and a local generative model to output a grounded answer based on the retrieved academic articles.

Furthermore, we achieved the second research objective by selecting the **granite-embedding:30m** as the dense retriever and **Qwen2.5:3B** as the most suitable generator using automatic evaluation metrics. The retriever was selected by referring to the Recall@n, while the generator was identified using its semantic consistency experiment and assessment of the refusal capability.

Finally, we optimized the selected RAG pipeline and evaluated its effectiveness using the RAGAS evaluation framework. The optimized pipeline improved the overall RAGAS Index, with the most notable improvement observed in **Faithfulness (19.3%),** while maintaining high Answer Relevancy and improving Context Recall.

Despite the proposed RAG pipeline demonstrating promising performance, this study still has several limitations. Firstly, the proposed pipeline is only workable for English-language academic articles, which limits the generalizability of the findings to multilingual literature. Also, our knowledge base is constructed using abstracts instead of the whole documents. While it can be the most cost-effective approach, but also causes loss of track on the details of the article. Finally, the retriever optimization process prioritizes improving the Context Recall, resulting in noticeable decrease in Context Precision. Although the trade-off is anticipated, it also introduces more relevant document chunks into the retrieved context.

Our future work may focus on extending the framework to support multilingual academic datasets and incorporating full-text research articles as the knowledge source. In addition, future studies could investigate retrieval strategies that better balance Context Recall and Context Precision, such as adaptive retrieval, dynamic context selection, or more advanced re-ranking techniques. These improvements could further enhance the effectiveness and robustness of locally deployed RAG systems for academic question answering.

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